Mixed Reality Navigation System for Ultrasound-Guided and Fusion Biopsies Using Markerless Instrument Tracking

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ABSTRACT

Percutaneous needle biopsy is commonly performed under ultrasound guidance to visualize the target lesion and monitor needle position. This study aims to improve spatial information for the operator by accurately visualizing the ultrasound image at its true anatomical location and scale within a head-mounted display, co-registered with pre-procedural imaging data. Current mixed reality systems rely on fiducial markers to track the ultrasound probe, which limits their practical utility in clinical settings; accordingly, this study evaluated a custom markerless method. By displaying the ultrasound image relative to the probe's spatial position, the system enables seamless fusion with pre-procedural imaging data. Combined with markerless needle tracking, this approach provides a complete navigation solution. The ultrasound probe tracking method achieved a refresh rate exceeding 10 Hz with accuracy suitable for clinical applications. Target registration error analysis between 2D ultrasound and 3D computed tomography demonstrated a 95% likelihood that the error remains within 3.41 mm. A biopsy navigation assessment test confirmed the system's strong potential to reduce the number of puncture attempts compared to conventional ultrasound guidance alone. By providing comprehensive spatial and imaging information, the navigation system may contribute to reduced operator error and improved procedural success rates.

Introduction

Biopsy is a medical procedure that involves removal of a tissue sample from a living individual for diagnostic examination[1]. One of its most commonly performed type is percutaneous needle biopsy (PNB), which is a minimally invasive method that relies on inserting a needle through the skin[2]. It is conducted with the assistance of imaging modalities like ultrasound (US), fluoroscopy, computed tomography (CT), magnetic resonance imaging (MRI), cone beam CT (CBCT) and positron emission tomography CT (PET-CT).

US is often preferred over other modalities because it lacks ionizing radiation and is portable, accessible, and fast, while providing real-time needle visualization and precise guidance in almost any anatomical plane[3,4]. It can also adjust to tissue changes such as pressure, respiration, injection, and bladder expansion. US-guided PNB is widely used for diagnosing thoracic lesions[5], peripheral pulmonary lesions[6] or musculoskeletal soft tissue masses[7]. Reported diagnostic success rates in past studies include 90.5% for thoracic lesion (lesion sizes 1.5-16 cm, mean 7 cm)[5], 81.8% for peripheral pulmonary lesions (72.0% $\leq$ 20 mm, 86.8% for 21–49 mm, 79.7% $\geq$ 50 mm; lower accuracy due to necrosis in 28.8% of cases)[6], and 81.4% for peripheral pulmonary nodules $\leq$ 2 cm[8].

Recent work has explored head-mounted display (HMD) applications for US-guided procedures. Nguyen et al.[9] have streamed and rendered US images in HMD with respect to the tracked probe by detecting specific markers printed on a custom probe case. The study revealed a 17% faster completion time for novices using the mixed reality system; however, for experts, the system resulted in a 5% slower performance, likely due to their familiarity with traditional methods. Costa et al.[10] used QR codes to track both US probe and needle within HMD, displaying live US video stream and trajectory from needle tip to the lesion. The study revealed longer procedure times but improved needle alignment and lesion targeting compared to a conventional approach. Li et al.[11] tracked retro-reflective spheres both on the US probe and needle with depth camera. An examination carried out on a phantom containing a tumor with a 5 mm radius showed that mixed reality guidance increased the procedure success rate from 26% to 98% for out-of-plane punctures and from 45% to 93% for in-plane punctures.

Fusion biopsy, which combines real-time images provided by the US and high-resolution features of MRI, is now widely used for prostate cancer diagnosis[12–14]. Studies report that fusion biopsy achieves higher overall diagnostic accuracy and sensitivity than conventional US-guided techniques, including detecting high-grade tumors (Gleason score $\geq$ 7)[15,16]. The

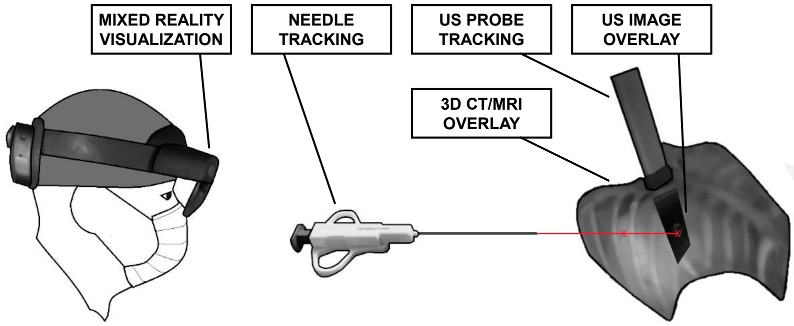

Figure 1. Scheme of mixed reality biopsy navigation system with US imaging. The pre-procedural CT/MRI data is reconstructed into a volumetric model that is displayed on HMD and aligned with patient's physical body. The US probe tracking enables the display of the US image in real size and position, correlated with the probe's location. The solution also enables the selection of a needle trajectory and guides the operator along this path during needle placement.

fusion of MRI and US is typically performed using three distinct methods. One approach is cognitive fusion, which relies on MRI to identify suspicious region and then target it using US without any specialized software[17,18]. This method depends heavily on operator experience and does not outperform standard biopsy techniques[19]. Another techniques are rigid registration that is based on identifying corresponding anatomical landmarks and computing translation and rotation to bring one image set into alignment with the other[20,21] and elastic registration that additionally tries to compensate for possible deformations, such as those caused by patient movements, pressure from surgical instruments, or injections[22]. Research conducted by Hale et al.[23] indicated that the rigid registration achieves a registration error of 4.87 ± 3.50 mm but elastic registration does not considerably improve the results, achieving error of 4.11 ± 2.09 mm.

The current research advances US applications in biopsy by building upon mixed reality navigation system introduced in[24]. It contained markerless needle tracking and imaging data superimposition onto the patient's physical body. The present work incorporates markerless US probe tracking, enabling real-time overlay of US images in accurate scale and anatomical location being examined. The system supports marking of target and injection sites using gestures and voice commands and guides the operator to precisely perform the procedure. This approach reduces human error, potentially shortens procedure time, and improve detection accuracy. Moreover it enables real-time fusion of 2D US with 3D CT/MRI without any complex intensity-based registration algorithms, extending fusion biopsy beyond prostate cancer. The fully markerless setup provides a better user experience and greater flexibility by eliminating the need to maintain focus on markers to ensure they remain in the camera's field of view.

Methods

The mixed reality biopsy navigation system consists of three major elements, presented in Fig. 1. It involves reconstructing CT/MRI data into a volumetric model that is visualized on HMD, along with needle and US probe tracking to overlay US images and assist the user during needle insertion.

Mixed reality visualization and patient registration

Visualization of imaging data in mixed reality is done using CarnaLife Holo technology (MedApp S.A., Poland)[25]. It processes DICOM data into a 3D volumetric model that can be visualized through HMD. Users can interact with it including moving, rotating, scaling, slicing, or taking measurements[26]. The system can also superimpose the visualized data onto the patient's body, maintaining accurate real-world dimensions[27,28]. This requires scanning the subject with radiological markers that serve as reference points and remain clearly visible in the imaging data across multiple slices. By matching corresponding points in the imaging data and physical sites within the world coordinate system of the headset as presented in Fig. 2, the transformation between these two point sets is computed. In a result, the imaging data is adjusted to real-world dimensions and perceived in HMD as a volumetric model aligned with the subject[24].

Markerless surgical instruments tracking

The system is embedded with simultaneous tracking of biopsy needle and US probe without the need for physical markers, as presented in Fig. 3. Needle tracking enables guiding the operator to precisely inject the needle at selected location, ensuring that it reaches the target area. Moreover, US probe tracking allows for real-time US images to be displayed relatively to the probe location, with an accurate scale and orientation.

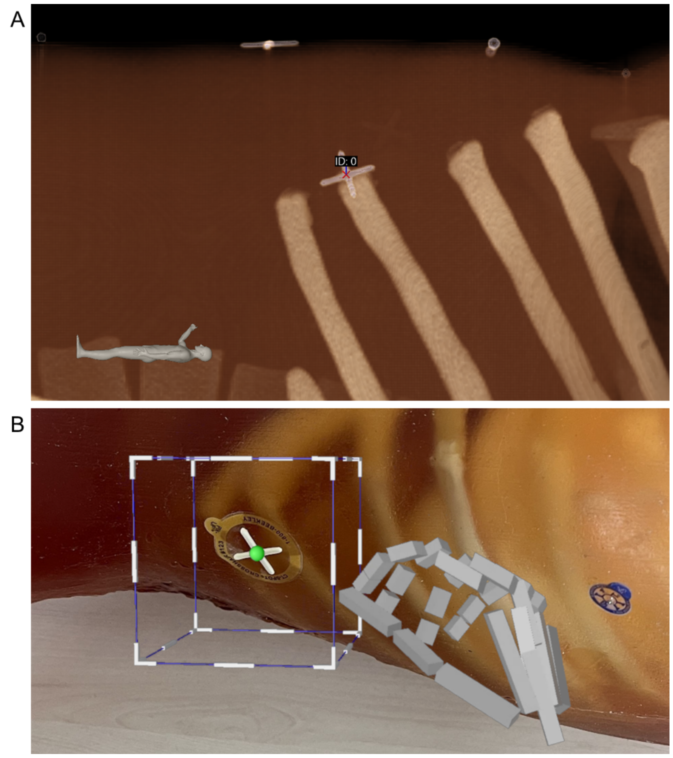

Figure 2. Selecting corresponding reference points in the imaging data and physical sites in world coordinate system of the headset to perform patient registration: (**a**) selected reference point in the imaging data; (**b**) mixed reality interface to select corresponding reference point within HMD on a physical site.

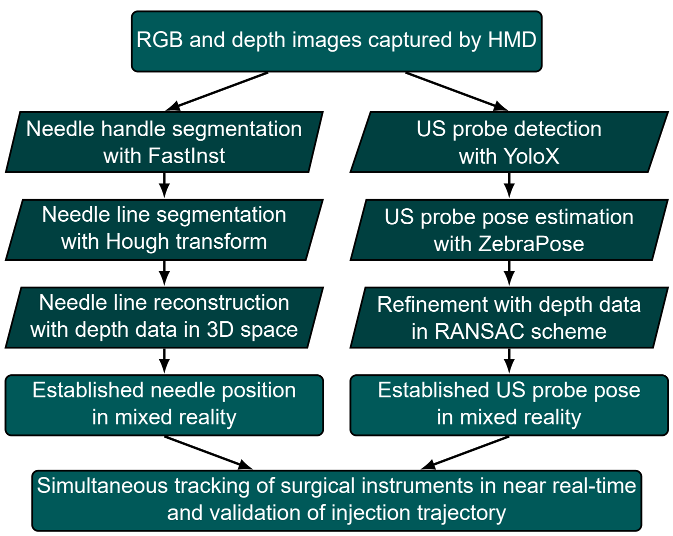

Figure 3. Scheme of markerless surgical instruments tracking pipeline. The inputs are RGB and depth images captured by the built-in HMD cameras and in result algorithm outputs estimated needle position and pose of US probe. It enables possibility to validate correctness of injection in predefined trajectory.

The system utilizes the Microsoft HoloLens 2 headset that is equipped with a set of built-in cameras and sensors accessible by Research Mode[29]. The process of needle tracking begins with segmenting the tool's handle on RGB image with FastInst[30], followed by identifying the proper needle line through the Hough transform. Subsequently, the tool's position within a mixed reality space is reconstructed by aligning points captured by depth camera with the predefined needle line[24].

US probe tracking begins by detecting the tool in RGB image with YoloX[31] and estimating its pose using ZebraPose[32].

The Perspective-n-Point (PnP) problem is solved using the Progressive-X (Prog-X) algorithm[33], which iteratively generates pose hypotheses, quickly rejects redundant ones and integrates the new ones into a compound set using energy minimization, enforcing spatial coherence by penalizing inconsistent or non-adjacent 3D–2D projections. This method has shown high efficiency on the LM-O dataset[34] that contains multiple occluded and texture-less objects[32]. The predicted pose is then refined with depth data by establishing correspondences between model point cloud and points captured by depth camera[35]. These correspondences are determined by projecting the model point cloud onto the RGB image plane, which is aligned with the depth camera image. Using the relationship between model points transformed by the predicted pose and their distance from camera we exclude those points that are not visible from the camera's view. Additionally, points not related to the object are excluded using segmented mask predicted by the ZebraPose which handles occlusions by other elements such as a hand. Finally, a rigid transformation between the two point sets is computed via least-squares fitting[36] in a RANSAC scheme, and pose stability is improved using Kalman filtering and Exponential Moving Average (EMA). The study utilized US Chison SonoAir 70 (Chison, China) system with linear probe.

Since the instrument tracking relies solely on the built-in sensors, no extra calibration step is required. The predicted pose is obtained within internal Photo-Video camera coordinate system and can be directly transformed to the world coordinate system of the headset. That is a major simplification over solutions that consists of external cameras or tracking systems which require calibration before each procedure.

Multi-modal image fusion

Unifying the coordinate systems of tracked surgical instruments, pre-procedural imaging data, and HMD enables the fusion of 2D US images with 3D CT/MRI data. To achieve this, the reconstructed volumetric model from the pre-procedural scan must first be accurately aligned with the patient's physical body. Once the US probe is tracked, real-time US images can be also overlaid on the patient's body in their true spatial position and scale, allowing seamless integration with the pre-procedural imaging data. The estimated position and orientation of the US image determines the cutting plane through the superimposed volumetric data, creating a cross-section of the 3D CT/MRI. Within this method it is possible to overlay 2D US onto 3D CT/MRI without any advanced intensity-based registration algorithms. As a result, the operator gains a comprehensive information about the patient's anatomy, enabling more precise and informed decision-making during the procedure.

Results

Ablation Study

The first step in markerless US probe tracking pipeline is to determine its pose from RGB image. The selection of a model for this task was guided by the need to balance high accuracy with minimal inference time given real-time performance requirements. The study compared ZebraPose against PVNet[37] in terms of US probe pose estimation from monocular RGB images, considering also the difference in accuracy between RANSAC and Prog-X in solving the PnP problem. Moreover, the study examined the influence of incorporating depth data by evaluating output poses after the refinement step and those predicted by the HiPose model[38]. The refinement was performed without utilizing the Kalman filter and EMA, since they necessitate input data from consecutive time steps. The assessment was conducted using Maximum Symmetry-Aware Surface Distance (MSSD) and Maximum Symmetry-Aware Projection Distance (MSPD) metrics[39] that accounts for both the translation and rotation of the object. The MSSD measures the maximal surface distance between the estimated and ground-truth object poses, while the MSPD measures the maximal pixel displacement between their projections:

$$e_{\mathrm{MSSD}}(\hat{P},\bar{P},\mathscr{S}_M,\mathscr{V}_M) = \min_{S\in\mathscr{S}_M}\max_{x\in\mathscr{V}_M}\left\|\hat{P}x-\bar{P}Sx\right\|_2 \tag{1}$$

$$e_{\mathrm{MSPD}}(\hat{P},\bar{P},\mathscr{S}_M,\mathscr{V}_M) = \min_{S\in\mathscr{S}_M}\max_{x\in\mathscr{V}_M}\left\|proj(\hat{P}x)-proj(\bar{P}Sx)\right\|_2 \tag{2}$$

where M is the object's 3D model, $\hat{P}$ is the estimated pose, $\bar{P}$ is the ground-truth pose, $\mathscr{S}_M$ is a set of object symmetries and $\mathscr{V}_M$ is a set of points sampled from the object's 3D model. Moreover, errors in estimating the object's origin position and projection were also studied. Both models were trained on almost 16000 images and tested on 2200 images gathered using Intel Realsense D455 and D435i stereo cameras (Intel, USA) with annotated US probe poses. The test set included mainly hand-held and occlusion scenarios. The image annotation process involved reconstructing the point cloud of a given scene and manually aligning the 3D model of the object with its corresponding location. This approach made it possible to determine the object's translation and rotation relative to the camera coordinate system for each image. The recalls for accurate estimates at various thresholds are shown in Fig. 4. The superior outcomes of ZebraPose over PVNet demonstrate that the

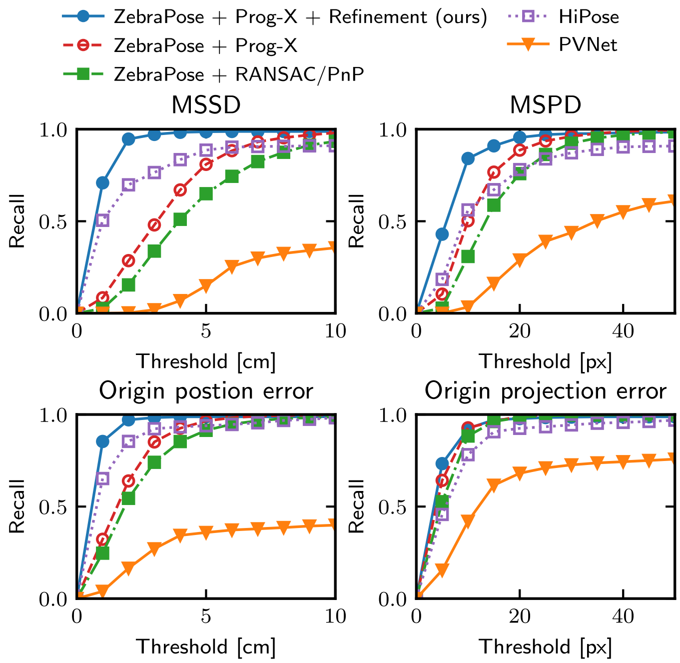

Figure 4. Recalls of US probe pose estimations in camera coordinates at various thresholds for evaluated models. Origin point is the geometric center of the object. ZebraPose with Prog-X solver excels with monocular input. Incorporating depth data significantly enhances accuracy, where refining ZebraPose output using our developed technique yields superior results compared to HiPose.

model excels at handling low-textured objects with limited number of distinctive features. The use of Prog-X instead of the standard RANSAC/PnP approach also increases the accuracy level. The refinement with depth data significantly improves the results: 70.86% of cases fall below the 1 cm MSSD threshold and 80.09% below the 10 px MSPD threshold, compared to only 8.41% and 50.27%, respectively, when using ZberaPose with Prog-X alone. An exemplary predicted poses before and after refinement are presented in Fig. 5. Overall, HiPose generally achieves pose estimations with less error compared to ZebraPose alone, with 50.50% of cases below a 1 cm MSSD threshold and 56.14% below a 10 px MSPD threshold, though it does not reach the precision achieved by refining ZebraPose's results with the developed method. However, there are certain instances where HiPose performs worse than ZebraPose, with only 91.00% of cases falling below the 10 cm MSSD threshold, as opposed to 98.05% for ZebraPose. The achieved results demonstrate that utilizing depth data is crucial for accurate pose estimation, especially in determining the correct distance of the object from the camera.

The impact of each step in the custom refinement method using depth data was assessed by estimating object orientation with different versions of the algorithm. Each subsequent version activated the next element, which was then compared to the reference orientation determined by the surrounding ArUco codes on a prepared 3D-printed plate. In this scenario, the probe remained stationary while the headset moved around it, observing the object from various perspectives. The results of 1000 consecutive iterations of the algorithms are presented in Table 1. In both non-occluded and hand-occluded scenarios, the error after all steps is over one-third of a baseline. Without occlusion, RANSAC doubles the accuracy, though the mask does not substantially affect the results. Nevertheless, for occlusion scenarios, RANSAC substantially enhances its influence on the results after employing the mask to remove points not related to the tracked object.

Comparison to Marker-based Tracker

Pose obtained with ArUco codes used in ablation study has limited precision as a mean registration accuracy of the marker center is above 1 mm even in the best case scenario[40]. To effectively validate positioning accuracy of the developed solution it was compared to a different navigation system. The study utilized ClaroNav MicronTracker H3-60 (ClaroNav, Canada) which is designed to detect and track objects by identifying passive, visible-light markers of unique black-and-white checkered patterns called Xpoints[41]. It has root mean square calibration accuracy of 0.20 mm, measured for single Xpoint over 20000 averaged positions at depths in range 40-100 cm.

To avoid discrepancies caused by coordinate system alignment between both systems, the study examined relative motion, specifically observing the change in the estimated pose after moving or rotating the object[42]. The HMD and MicronTracker

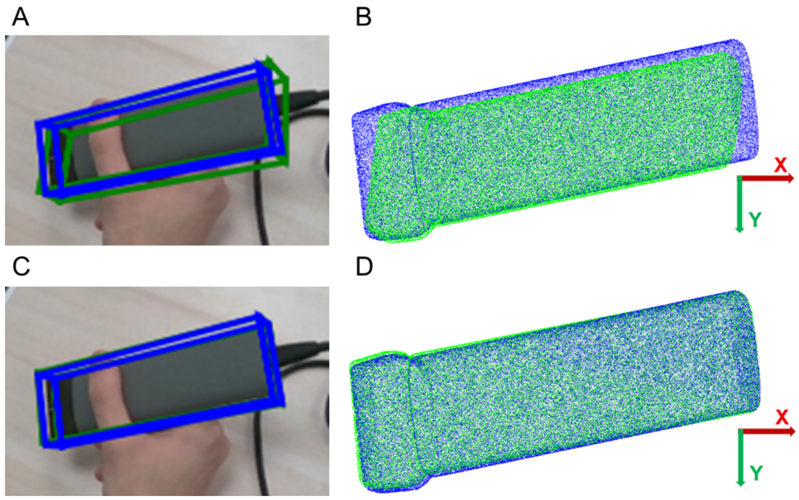

Figure 5. ZebraPose predictions using Prog-X solver before and after refinement with depth data where blue boxes/points represents ground-truth poses, green represents predictions: (**a**) MSPD 9.79 px; (**b**) MSSD 5.14 cm; (**c**) MSPD 2.25 px; (**d**) MSSD 0.31 cm.

Table 1. Impact of post-processing elements to refine the output pose in world coordinates. The error is calculated as a mean of angles between corresponding direction vectors of the estimated and ground-truth orientations. The baseline version of the algorithm performs least squares fitting across all model and depth points. The RANSAC-based approach selects the transformation achieving the highest number of inliers. The predicted object mask is then used to perform early outlier rejection. Finally, the pose is stabilized using a combination of a Kalman filter and EMA.

Post-processing	Non-occluded	Hand-occluded
Baseline	10.79 ± 5.42°	12.94 ± 6.47°
+ RANSAC	4.87 ± 3.03°	9.81°± 3.89°
+ Mask	4.20 ± 2.77°	4.86 ± 3.03°
+ Kalman filter	3.51 ± 2.18°	3.85 ± 2.16°
+ EMA	2.94 ± 1.60°	3.21 ± 1.69°

were positioned statically in fixed locations, each observing the object from the opposing directions. For translation error measurements, the US probe was translated by 5 cm up to 30 cm, the HMD was placed 60 cm from the object center at the midpoint, angled at 50° towards the object while MicronTracker was placed 65 cm from the marker at the midpoint and angled at 15° towards it. In case of rotation error measurements, the US probe was rotated by 10° up to 90°, the HMD was placed 45 cm from the object, angled at 50° towards it while MicronTracker was placed 65 cm from the rotation center and angled at 35° towards it. For both tests the initial object pose estimated by the proposed solution was averaged over 20 consecutive iterations and for each step 20 consecutive estimations were compared with the readings from MicronTracker that served as the ground-truth. In case of rotation test, the delta was calculated along two axes perpendicular to the axis of rotation and taken the average of them. The results of the tests are presented in Fig. 6. They demonstrate a reliable positioning accuracy that aligns closely with the readings from the ground-truth.

Runtime analysis

The sensor data is transmitted from the HMD to external computer, where complete processing takes place. The visualization is also rendered on the external machine and streamed to the HMD simultaneously, ensuring immediate updates and smooth end-to-end refresh rates. The runtime test was conducted on a machine equipped with an AMD Ryzen 9 5900HX CPU and an Nvidia RTX3080 GPU. The algorithm was implemented in C++ with Prog-X optimized to enable real-time performance and models accelerated with TensorRT. The measured runtimes are averaged over 1000 iterations. The data transmission takes 19.47 ± 2.44 ms, detection with YoloX takes 18.73 ± 0.67 ms, building 2D-3D correspondences by ZebraPose takes 11.46 ± 0.78 ms, Prog-X runs at 28.04 ± 6.25 ms and refinement of the predicted pose with depth data takes 12.61 ± 0.85 ms. The total duration required to update the object's pose is 90.31 ± 6.84 ms, resulting in end-to-end refresh rate over 10 Hz.

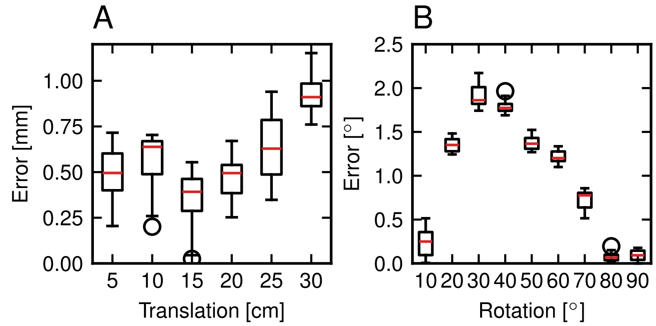

Figure 6. Results of the relative motion test: (**a**) translation error; (**b**) rotation error. Estimations of the markerless solution closely aligns with readings from the marker-based tracker.

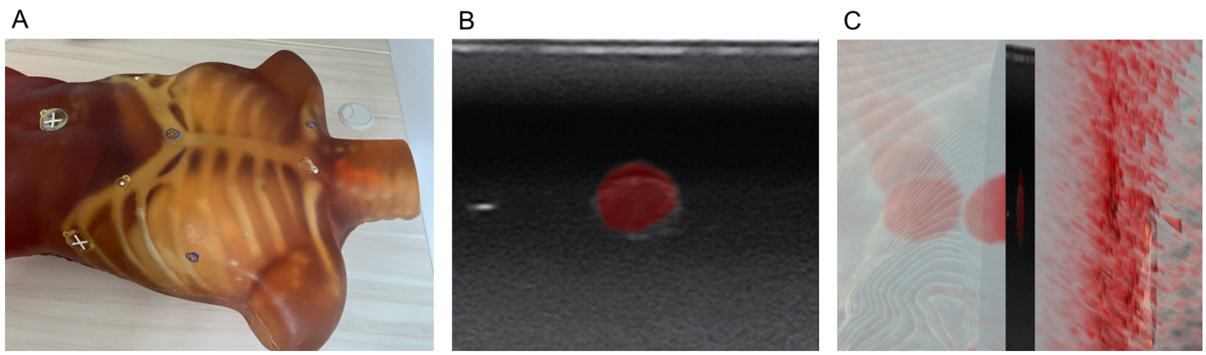

Figure 7. Rigid 2D US - 3D CT fusion (**a**) Rigid radiological phantom used for the evaluation (**b**) US image with red overlay representing lesion region in CT data of mean TRE 1.28 ± 0.39 mm; (**c**) the same fused data shown from a different viewing angle, with the 2D US image positioned within the 3D CT volume.

CT-US Target Fusion Error

Evaluation of the target registration error (TRE) between CT and US was carried out using a radiological phantom with embedded lesion imitations that was subjected to CT scanning. Twenty samples of US images with estimated positions and orientations within 3D CT volumetric model were collected from various locations. Each sample underwent 20 measurements between corresponding points of the lesion contours on the US image and 3D CT cross-section. An example of performed fusion is depicted in Fig. 7. The mean TRE across all measurements was found to be 1.97 ± 0.89 mm, and the complete distribution of errors is illustrated in Fig. 8. The survival function analysis revealed a 0.95 likelihood that the error remains within 3.41 mm. It should be noted that although the test was performed using CT imaging, the procedure and the user interface remain the same in the case of MRI.

Biopsy Navigation Assessment

The impact of utilizing developed mixed reality navigation with markerless surgical instruments tracking was validated by simulating biopsy procedure on a pork shank meat from a butcher store. The test did not incorporate fusion of US with CT/MRI but evaluated in practice suitability of both needle and US probe tracking accuracy. For the experiment, one wooden ball of a given size (diameter 8–20 mm, in 2 mm steps) was inserted into the meat at a time. For each size, 10 injections were performed using both mixed reality navigation and US only. After each set, the ball was replaced with the next smaller size. The US was used to visualize the ball position in the meat and to confirm whether the needle successfully hit the target. Each attempt was performed from different sites, with injection depths ranging from 7 to 11 cm.

In order to guide the user to perform a precise needle injection, the target and puncture sites had to be first be selected. To select the target site, the user positioned both index finger tips to define a line that intersects the US image plane at the desired location. The puncture site was selected with single index finder tip. The selection and confirmation of points began with the use of appropriate voice commands. The two chosen points defined a trajectory that the needle should follow during insertion. Through needle tracking, the operator was guided to maintain the designated path[24]. Mixed reality navigation interface is presented in Fig. 9.

The test was performed by a person without prior experience in performing medical procedures. The results are presented

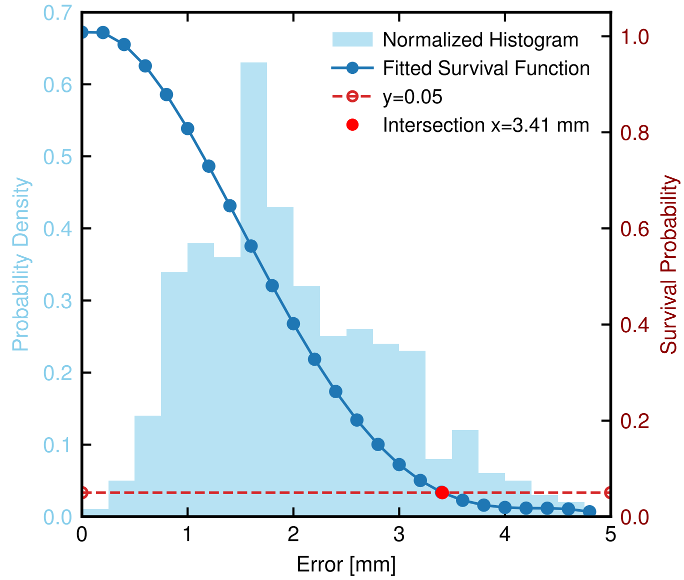

Figure 8. Distribution of target registration errors for 400 measurements. Survival function was fitted with 4th degree polynomial. The analysis indicates 0.95 likelihood that the error lies within 3.41 mm.

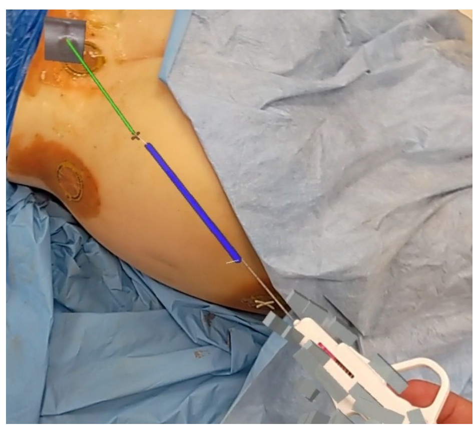

Figure 9. Interface of mixed reality navigation with US-guidance. The overlaid US image contains lesion region, allowing the system to designate target and injection points, guiding the user to align the estimated needle position (blue line) with the chosen trajectory (green line). The user task is to minimize the length of leading lines (white lines) that indicate the needle deviation from the trajectory.

in Fig. 10. The target hit rate was calculated as the ratio of the number of successful punctures when needle has reached the ball to the total number of attempts for each ball size. The experiment demonstrated that utilizing mixed reality navigation significantly improves procedural accuracy, as the operator successfully reached the target in 87.14% of cases compared to only 32.86% when using US alone.

Discussion

The study validated the use of markerless US probe tracking in a mixed reality system for biopsy support. Applying a deep learning model to this task is challenging, as it must balance high accuracy and low inference time. According to the authors of ZebraPose, recognized as one of the fastest models for 6D pose estimation of seen objects with reasonably high precision, the model still operates in 250 ms[32]. Performed optimizations enabled real-time operation and the applied post-processing aligned the model output with real-world data captured by depth camera. Compared to a well-established surgical tracking technology ClaroNav MicronTracker, the achieved accuracy proved suitable for clinical use.

Some recent studies have explored the markerless object pose estimation for HMD applications using only built-in sensors.

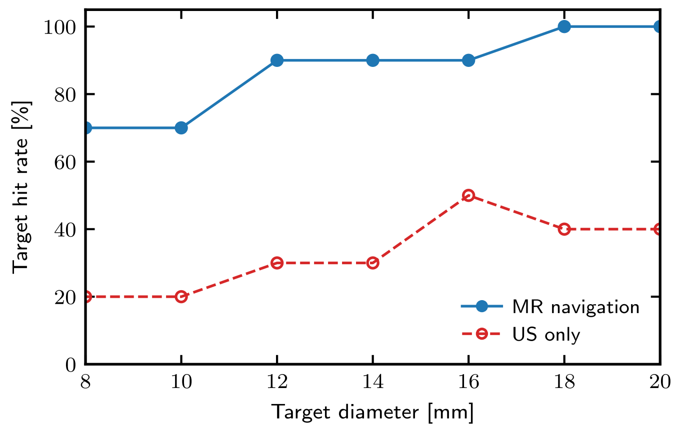

Figure 10. Mixed reality navigation assessment performed on the meat for injections at depths in range 7-11 cm. The use of developed solution outperforms standard US-guidance in terms of target hit rate.

Pöllabauer et al.[43] used a deep learning model to estimate 6D poses only from RGB images and anchor virtual overlays, reporting MSSD and MSPD values suitable for assistance but insufficient for high-precision alignment. Ali et al.[44] used Azure Object Anchors, trained on a 3D object model and relying solely on depth data, but noted limitations for similarly shaped or planar objects and for targets smaller than 70 cm. In contrast, our study combines RGB and depth data to take advantage of both modalities. RGB images provide color and texture information and are analyzed by deep learning models while depth camera data provides information about actual geometry and localization of the object. The employed framework can be applied to any object, although the resolution and quality of the built-in Time-of-Flight (ToF) camera might limit its applicability, which requires additional research.

Multi-modal fusion of US and CT/MRI is crucial in various clinical contexts, as the advantages of each modality enhance diagnosis and procedure planning. A substantial amount of previous research focused on improving this process and handling its major difficulties. Gueziri et al.[45] proposed a framework for rigid registration between CT and intraoperative US of a single vertebra, achieving a median accuracy of 1.48 mm and requiring 11 seconds of computation time. Wei et al.[46] employed a classification network to estimate the US probe orientation, then refined through segmentation and achieved a mean registration error of 4.7 ± 1.7 mm. Chi et al.[47] proposed a method based on deep learning for registering 2D US with 3D CT for kidney scans of free breathing, achieved a mean contour distance of 0.79 ± 0.22 mm. Last but not least Wang et al.[48] proposed a solution to register 2D US with 3D MRI of liver images, obtaining mean TRE 4.95 ± 2.23 mm. The literature often highlights that a major challenge in registration involves US noise. The presented solution relies only on estimated US probe pose instead of complex intensity-based registration algorithms, yet it still delivers satisfactory performance, making it simple and adaptable across many anatomies and procedures. The user can perform real-time fusion of images as the results can be observed immediately. However, one limitation is the requirement for accurate 3D CT/MRI alignment with the patient, which is currently performed by manually identifying landmarks. Nonetheless, recent research in this field may lead to automate this process without the need for markers[49]. Another study constraint is performing the fusion on a rigid, non-deformable phantom, which limits the ability to simulate US examination on human participants. Further research should also consider the potential misalignment between the 2D US plane and the 3D CT/MRI cross-section, although obtaining ground-truth data remains a challenge.

The standard US-guided biopsy lack of spatial information to coordinate needle movement in relation to the lesion position, especially for punctures out of the image plane. The developed system addresses this issue as the user perceive anchored US image with marked target and is guided to adjust proper needle placement. The conducted biopsy navigation assessment test proved usability of the solution as the target hit rate increased by over 2.5 times compared to using US alone. This directly affects the reduction of required punctures, thereby decreasing complication rates, blood loss, and patient discomfort.

One of the study limitations is relying only on the built-in ToF camera for depth measurements that is prone to errors caused by distractions such as lighting, reflections, colors and materials[50,51]. The further work should explore the use of HMDs equipped with stereo cameras as they provide higher resolution and a limited amount of noise[52]. The evaluation of the developed markerless US probe tracking pipeline on data captured with an Intel RealSense camera, as reported in the ablation study, demonstrated the portability of the solution. Nevertheless, generating precise disparity maps without compromising real-time performance continues to be a demanding task[53].

Conclusion

The study presented and evaluated mixed reality navigation system for biopsy that features fusion between 2D US and 3D CT/MRI. The developed markerless surgical instruments tracking method was validated against state-of-the-art RGB-only and RGB-D methods and achieved the best outcomes in case of US probe pose estimation. The method was also evaluated alongside a different navigation system, and the results were closely matched. The CT-US target registration error test indicated a 0.95 likelihood that the error remains within 3.41 mm which is suitable for clinical use. The biopsy navigation assessment test demonstrated that mixed reality navigation significantly enhances procedural success while minimizing human errors.

Data availability

The datasets created and analyzed during the current study are not publicly available due to privacy and confidentiality restrictions but are available from the corresponding author on reasonable request.

References

1. Anjum, R. *et al.* Biopsy techniques: A review. *J. Oral Medicine, Oral Surgery, Oral Pathol. Oral Radiol.* **10**, 181–184, DOI: 10.18231/j.jooo.2024.034 (2024).
2. Veltri, A., Bargellini, I., Giorgi, L., Almeida, P. A. M. S. & Akhan, O. CIRSE Guidelines on Percutaneous Needle Biopsy (PNB). *Cardiovasc. Interv. Radiol.* **40**, 1501–1513, DOI: 10.1007/s00270-017-1658-5 (2017).
3. Serati, L. *et al.* Diagnostic yield and accuracy of image-guided percutaneous core needle biopsy of paediatric solid tumours: An experience from Italy. *Pediatr. Hematol. Oncol. J.* **6**, 12–17, DOI: 10.1016/j.phoj.2020.10.003 (2021).
4. Kim, J. W. & Shin, S. S. Ultrasound-Guided Percutaneous Core Needle Biopsy of Abdominal Viscera: Tips to Ensure Safe and Effective Biopsy. *Korean J. Radiol.* **18**, 309–322, DOI: 10.3348/kjr.2017.18.2.309 (2017).
5. Portela-Oliveira, E. *et al.* Ultrasound-guided percutaneous biopsy of thoracic lesions: High diagnostic yield and low complication rate. *Clin. Radiol.* **76**, 281–286, DOI: 10.1016/j.crad.2020.12.004 (2021).
6. Guo, Y.-Q. *et al.* Ultrasound-Guided Percutaneous Needle Biopsy for Peripheral Pulmonary Lesions: Diagnostic Accuracy and Influencing Factors. *Ultrasound Medicine Biol.* **44**, 1003–1011, DOI: 10.1016/j.ultrasmedbio.2018.01.016 (2018).
7. Szaro, P., Wong, A., Blain, E., Gataa, K. G. & Geijer, M. The diagnostic value of ultrasound-guided percutaneous core needle biopsy of musculoskeletal soft tissue lesions. *J. Ultrason.* **21**, 22–33, DOI: 10.15557/JoU.2021.0004 (2021).
8. Huang, W. *et al.* Ultrasound-Guided Percutaneous Core Needle Biopsy of Peripheral Pulmonary Nodules $\leq$ 2 cm: Diagnostic Performance, Safety and Influence Factors. *Front. Oncol.* **11**, 671884, DOI: 10.3389/fonc.2021.671884 (2021).
9. Nguyen, T., Plishker, W., Matisoff, A., Sharma, K. & Shekhar, R. HoloUS: Augmented reality visualization of live ultrasound images using HoloLens for ultrasound-guided procedures. *Int. J. Comput. Assist. Radiol. Surg.* **17**, 385–391, DOI: 10.1007/s11548-021-02526-7 (2022).
10. Costa, N. *et al.* Augmented Reality-Assisted Ultrasound Breast Biopsy. *Sensors* **23**, 1838, DOI: 10.3390/s23041838 (2023).
11. Li, H. *et al.* Navigate biopsy with ultrasound under augmented reality device: Towards higher system performance. *Comput. Biol. Medicine* **174**, 108453, DOI: 10.1016/j.compbiomed.2024.108453 (2024).
12. Siddiqui, M. M. *et al.* Comparison of MR/Ultrasound Fusion–Guided Biopsy With Ultrasound-Guided Biopsy for the Diagnosis of Prostate Cancer. *JAMA* **313**, 390–397, DOI: 10.1001/jama.2014.17942 (2015).
13. Bak, S., Budzen, K., Wesolowski, P., Stachura, T. & Gladysz, K. Review of diagnostic methods for prostate cancer with consideration of MRI-TRUS fusion biopsy. *J. Educ. Heal. Sport* **12**, 720–728, DOI: 10.12775/JEHS.2022.12.09.085 (2022).
14. Musaddaq, T. & Musaddaq, B. Recent Advances in Image-Guided Tissue Sampling. *Cureus* **16**, e71613, DOI: 10.7759/cureus.71613 (2024).
15. Fang, J.-h. *et al.* Comparative diagnostic accuracy of multiparametric magnetic resonance imaging-ultrasound fusion-guided biopsy versus systematic biopsy for clinically significant prostate cancer. *PeerJ* **11**, e16614, DOI: 10.7717/peerj.16614 (2023).
16. Zhu, G. & Wang, Q. Comparisons between magnetic resonance/ultrasound fusion-guided biopsy and standard biopsy in the diagnosis of prostate cancer: A prospective cohort study. *Medicine* **97**, e11962, DOI: 10.1097/MD.0000000000011962 (2018).

17. Chang, S. D. *et al.* MRI Targeted Prostate Biopsy Techniques: AJR Expert Panel Narrative Review. *Am. J. Roentgenol.* **217**, 1263–1281, DOI: 10.2214/AJR.21.26154 (2021).

18. Marks, L., Young, S. & Natarajan, S. MRI–ultrasound fusion for guidance of targeted prostate biopsy. *Curr. Opin. Urol.* **23**, 43–50, DOI: 10.1097/MOU.0b013e32835ad3ee (2013).

19. Şahin, B., Dörücü, D., Tinay, İ., Filinte, D. & Çam, H. K. Is Cognitive MR Fusion Biopsy Superior to Standard TRUS Guided Prostate Biopsy? Our Clinical Experience. *Bull. Urooncology* **21**, 35–39, DOI: 10.4274/uob.galenos.2021.2021.2.4 (2022).

20. Naselli, A. *et al.* Fusion prostate biopsy: Tips and tricks to improve rigid registration. *J. Men's Heal.* **18**, 1, DOI: 10.31083/j.jomh1805120 (2022).

21. Cornud, F. *et al.* TRUS–MRI image registration: A paradigm shift in the diagnosis of significant prostate cancer. *Abdom. Imaging* **38**, 1447–1463, DOI: 10.1007/s00261-013-0018-4 (2013).

22. Venderink, W., de Rooij, M., Sedelaar, J. P. M., Huisman, H. J. & Fütterer, J. J. Elastic Versus Rigid Image Registration in Magnetic Resonance Imaging–transrectal Ultrasound Fusion Prostate Biopsy: A Systematic Review and Meta-analysis. *Eur. Urol. Focus.* **4**, 219–227, DOI: 10.1016/j.euf.2016.07.003 (2018).

23. Hale, G. R. *et al.* Comparison of Elastic and Rigid Registration during Magnetic Resonance Imaging/Ultrasound Fusion-Guided Prostate Biopsy: A Multi-Operator Phantom Study. *The J. Urol.* **200**, 1114–1121, DOI: 10.1016/j.juro.2018.06.028 (2018).

24. Trojak, M., Stanuch, M., Kurzyna, M., Darocha, S. & Skalski, A. Mixed Reality Biopsy Navigation System Utilizing Markerless Needle Tracking and Imaging Data Superimposition. *Cancers* **16**, 1894, DOI: 10.3390/cancers16101894 (2024).

25. Wierzbicki, R. *et al.* 3D mixed-reality visualization of medical imaging data as a supporting tool for innovative, minimally invasive surgery for gastrointestinal tumors and systemic treatment as a new path in personalized treatment of advanced cancer diseases. *J. Cancer Res. Clin. Oncol.* **148**, 237–243, DOI: 10.1007/s00432-021-03680-w (2022).

26. Tsai, T.-Y. *et al.* Feasibility and accuracy of real-time 3D-holographic graft length measurements. *Eur. Hear. J. - Digit. Heal.* **5**, 101–104, DOI: 10.1093/ehjdh/ztad071 (2024).

27. Bronowicki, K. *et al.* An attempt to evaluate the use of mixed reality in surgically treated pediatric oncology patients. *npj Digit. Medicine* **8**, 262, DOI: 10.1038/s41746-025-01638-7 (2025).

28. Galyfos, G. *et al.* Using a novel three-dimensional holographic technology to perform open vascular surgery procedures. *J. Vasc. Surg. Cases, Innov. Tech.* **10**, 101440, DOI: 10.1016/j.jvscit.2024.101440 (2024).

29. Ungureanu, D. *et al.* HoloLens 2 Research Mode as a Tool for Computer Vision Research. *arXiv preprint arXiv:2008.11239* DOI: 10.48550/arXiv.2008.11239 (2020).

30. He, J., Li, P., Geng, Y. & Xie, X. FastInst: A Simple Query-Based Model for Real-Time Instance Segmentation. *Proc. IEEE/CVF Conference on Computer Vision Pattern Recognition* 23663–23672, DOI: 10.1109/CVPR52729.2023.02266 (2023).

31. Ge, Z., Liu, S., Wang, F., Li, Z. & Sun, J. YOLOX: Exceeding YOLO Series in 2021. *arXiv preprint arXiv:2107.08430* DOI: 10.48550/arXiv.2107.08430 (2021).

32. Su, Y. *et al.* ZebraPose: Coarse To Fine Surface Encoding for 6DoF Object Pose Estimation. *Proc. IEEE/CVF Conference on Computer Vision Pattern Recognition* 6738–6748, DOI: 10.1109/CVPR52688.2022.00662 (2022).

33. Barath, D. & Matas, J. Progressive-x: Efficient, anytime, multi-model fitting algorithm. *Proc. IEEE/CVF Int. Conf. on Comput. Vis. (ICCV)* (2019).

34. Brachmann, E. *et al.* Learning 6D Object Pose Estimation Using 3D Object Coordinates. *Eur. conference on computer vision* 536–551, DOI: 10.1007/978-3-319-10605-2_35 (2014).

35. Trojak, M., Jurgas, A., Stanuch, M., Kownacki, Ł. & Skalski, A. Mixed Reality 6D Object Pose Estimation using Deep Learning for Visual Markerless Surgical Navigation. *2025 IEEE Conference on Virtual Reality 3D User Interfaces Abstracts Workshops (VRW)* 947–952, DOI: 10.1109/VRW66409.2025.00193 (2025).

36. Arun, K. S., Huang, T. S. & Blostein, S. D. Least-Squares Fitting of Two 3-D Point Sets. *IEEE Transactions on Pattern Analysis Mach. Intell.* **PAMI-9**, 698–700, DOI: 10.1109/TPAMI.1987.4767965 (1987).

37. Peng, S., Liu, Y., Huang, Q., Zhou, X. & Bao, H. PVNet: Pixel-wise voting network for 6dof pose estimation. *2019 IEEE/CVF Conf. on Comput. Vis. Pattern Recognit. (CVPR)* 4556–4565, DOI: 10.1109/CVPR.2019.00469 (2019).

38. Lin, Y. *et al.* HiPose: Hierarchical Binary Surface Encoding and Correspondence Pruning for RGB-D 6DoF Object Pose Estimation. *Proc. IEEE/CVF Conference on Computer Vision Pattern Recognition* 10148–10158, DOI: 10.1109/CVPR52733.2024.00967 (2024).

39. Burde, V., Benbihi, A., Burget, P. & Sattler, T. Comparative Evaluation of 3D Reconstruction Methods for Object Pose Estimation. *2025 IEEE/CVF Winter Conference on Applications Computer Vision (WACV)* 7669–7681, DOI: 10.1109/WACV61041.2025.00745 (2025).

40. Florkowska, A. *et al.* Metrological Analysis of HoloLens 2 for Visual Marker-Based Surgical Navigation. *2023 IEEE International Conference on Imaging Systems Techniques (IST)* 1–6, DOI: 10.1109/IST59124.2023.10355714 (2023).

41. Liu, H.-G. *et al.* A Bulk Retrospective Study of Robot-Assisted Stereotactic Biopsies of Intracranial Lesions Guided by Videometric Tracker. *Front. Neurol.* **12**, 682733, DOI: 10.3389/fneur.2021.682733 (2021).

42. Eastman, R. D., Marvel, J., Falco, J. & Hong, T. Measurement science for 6DOF object pose ground truth. *2013 IEEE International Symposium on Robotic Sensors Environments (ROSE)* 196–201, DOI: 10.1109/ROSE.2013.6698442 (2013).

43. Pöllabauer, T., Rücker, F., Franek, A. & Gorschlüter, F. Detection and Pose Estimation of Flat, Texture-Less Industry Objects on HoloLens Using Synthetic Training. *Image Analysis* 569–585, DOI: 10.1007/978-3-031-31438-4_37 (2023).

44. Ali, M. H., Damouni, R., Nassar, A., Shimshoni, I. & Szpiro, S. F. A. Feasibility of Real Time 3D Object Detection for Accessibility Use Cases. *2025 IEEE Conference on Virtual Reality 3D User Interfaces Abstracts Workshops (VRW)* 1178–1179, DOI: 10.1109/VRW66409.2025.00235 (2025).

45. Gueziri, H.-E., Drouin, S., Yan, C. X. B. & Collins, D. L. Toward real-time rigid registration of intra-operative ultrasound with preoperative CT images for lumbar spinal fusion surgery. *Int. J. Comput. Assist. Radiol. Surg.* **14**, 1933–1943, DOI: 10.1007/s11548-019-02020-1 (2019).

46. Wei, W., Haishan, X., Alpers, J., Rak, M. & Hansen, C. A deep learning approach for 2D ultrasound and 3D CT/MR image registration in liver tumor ablation. *Comput. Methods Programs Biomed.* **206**, 106117, DOI: 10.1016/j.cmpb.2021.106117 (2021).

47. Chi, Y. *et al.* A two-step deep learning method for 3DCT-2DUS kidney registration during breathing. *Sci. Reports* **13**, 12846, DOI: 10.1038/s41598-023-40133-5 (2023).

48. Wang, Y. *et al.* Multimodal registration of ultrasound and MR images using weighted self-similarity structure vector. *Comput. Biol. Medicine* **155**, 106661, DOI: 10.1016/j.compbiomed.2023.106661 (2023).

49. Jurgas, A., Trojak, M., Stanuch, M. & Skalski, A. Markerless Registration and Visualization of Volumetric Medical Data for Enhanced Surgical Precision Using Head-Mounted Displays. *2025 IEEE Conference on Virtual Reality 3D User Interfaces Abstracts Workshops (VRW)* 925–930, DOI: 10.1109/VRW66409.2025.00189 (2025).

50. He, Y., Liang, B., Zou, Y., He, J. & Yang, J. Depth Errors Analysis and Correction for Time-of-Flight (ToF) Cameras. *Sensors* **17**, 92, DOI: 10.3390/s17010092 (2017).

51. Schuon, S., Theobalt, C., Davis, J. & Thrun, S. High-quality scanning using time-of-flight depth superresolution. *2008 IEEE Computer Society Conference on Computer Vision Pattern Recognition Workshops* 1–7, DOI: 10.1109/CVPRW.2008.4563171 (2008).

52. Agresti, G., Minto, L., Marin, G. & Zanuttigh, P. Deep learning for confidence information in stereo and tof data fusion. *Proc. IEEE International Conference on Computer Vision Workshops* 697–705, DOI: 10.1109/ICCVW.2017.88 (2017).

53. Tosi, F., Bartolomei, L. & Poggi, M. A Survey on Deep Stereo Matching in the Twenties. *Int. J. Comput. Vis.* **133**, 4245–4276, DOI: 10.1007/s11263-024-02331-0 (2025).

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Author contributions statement

Conceptualization: M.T., M.S. and A.S.; methodology: M.T., M.S. and A.S.; software: M.T.; formal analysis and investigation: M.T.; resources: A.S.; writing—original draft preparation: M.T.; writing—review and editing: M.T., M.S. and A.S.; visualization: M.T.; supervision: A.S.; funding acquisition: A.S.

Competing interests

The authors are MedApp S.A. employees and MedApp S.A. is the company that produces the CarnaLife Holo solution which was used for data visualization.